Fiscal Year:	FY 2015	**Task Last Updated:**	FY 12/10/2014
PI Name:	Vickers, Zata Ph.D.		
Project Title:	Factors Contributing to Food Acceptability and Consumption, Mood, and Stress on Long-term Space Missions		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	55108-1038	**Congressional District:**	4
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Crew Health NNJ10ZSA003N
Start Date:	02/15/2012	**End Date:**	02/14/2016
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No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Douglas, Grace	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: Extended to 2/14/2016 per NSSC information (Ed., 2/19/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mann, Traci (University of Minnesota) Redden, Joseph (University of Minnesota)		
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The three inter-related foci in this proposal explore factors that contribute to the long-term acceptability of a diet, in terms of satisfaction with and sufficient consumption of the foods, as well as associated moods and stress levels. The objective of our first focus is to examine behaviors related to preparing and consuming meals.

Understanding the relation between preparation effort and the acceptability and consumption of the prepared food, as well as effects on moods and stress, will inform decisions on the amount of time and types of activities crewmembers should allot to meal preparation. Understanding the relations between communal dining and special occasion meals and moods and stress will allow recommendations for dining behaviors that maximize positive moods and minimize stress.

To do this we will separately manipulate food preparation effort, food preparation creativity, and the number of people

Task Description:	eating together. We will measure the fluctuation in participants' moods and stress levels using questionnaires and physiological measurements. Meal and food acceptability will be measured using a variety of subjective rating scales. The objective of our second focus is to assess whether comfort foods repair individuals' moods, and if so, whether comfort foods are more effective than control foods at repairing moods. Understanding the effects of specific foods on crewmembers' food consumption, stress, and moods will inform decisions on the quantities of such mood-improving foods that should be made available on long missions. Participants first complete an online questionnaire about their food preferences (session one), then they come to the lab for two separate sessions in which we induce a negative mood and then give subjects a food to eat. We measure their mood after the negative mood induction, and then again after they eat the food, to assess whether the food led to improvements in their mood. The objective of our third focus is to curb boredom with repeatedly eaten foods by increasing appreciation for variety in one's diet without changing the available food items. Because the most unpleasant mood/emotion associated with food is boredom, and because boredom is highly related with a perceived lack of variety, understanding the conditions influencing people's perceptions and satisfaction with the variety in a diet will enable the crafting of practices that positively affect the mood and satisfaction of crewmembers with the foods on extended missions. Subjects in this study participate in three phases: phase I identified a liked food for each participant; phase II satiated participants on that single food; and phase III tested whether recalling variety reduced satiety and increased subsequent intake. For the first two focuses, positive results in ground based studies will be tested in flight definition studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our investigation of the role of food preparation and the eating situation on meal satisfaction, consumption, moods, and stress is highly related to these same issues on Earth. Is the selection and preparation of food 'therapy' for relieving stress? Or does it add to stress? And does this depend on specific characteristics of the individual? Information in the popular press strongly suggests that some foods (e.g., comfort foods) can relieve stress and repair bad moods. Does this really work on Earth? Are 'comfort foods' more effective at reducing stress and repairing moods than other foods? Than other activities? Our investigation of perceived variety is relevant for Earth-based situations where the variety of food people have available is limited or for people on highly restricted diets.
Task Progress:	Overview. Our third year goals focused primarily on preparation for flight studies on the International Space Station (Studies 2 & 3 from Focus 1 and study 6 from Focus 2). Focus 1. The role of food preparation and the eating situation on meal satisfaction, consumption, mood, and stress. Goals. The goals of this set of studies were to determine whether preparing food and/or having the ability to choose the ingredients in the food would improve mood and reduce stress. Study 1 (ground study): The main objectives of this study were to determine whether choosing meal components and/or preparing a meal would improve mood and reduce stress. We completed this study in 2012. The manuscript we submitted to Food Quality and Preference has been accepted for publication. Study 2 (Flight study, also known as Part A of Astro-Palate): The objective of this study is to determine the role of menu choice on meal satisfaction. This will be conducted once relatively earlier and then again later in the flight. Study 3 (Flight study, also known as Part B of Astro-Palate): The objective of this study is to determine the effects of eating situation on mood. Crewmembers will complete questionnaires before and after the meals. This will be done relatively earlier and then again later in the flight. Our first subject arrived at the International Space Station (ISS) in 2014. Other subjects have agreed to participate and are in varying stages of completing the pre-flight questionnaires. Next Steps. We will continue to recruit astronaut participants for this study and will continue working with the ISS Medical Project (ISSMP) team to implement the studies. Focus 2: The effects of specific food items on consumption, mood, and stress. Goals. The goals of this set of studies were to assess whether comfort foods repair individuals' mood, and if so, whether comfort foods are more effective at repairing mood than control foods or no foods. Studies 4 &5 (Ground studies). The manuscript we have prepared reporting this study has been accepted into the journal Health Psychology. Wagner, H. S., Ahlstrom, B., Redden, J. P., Vickers, Z., & Mann, T. (2014). The Myth of Comfort Food. Health Psychology, 33(12), 1552-1557. http://dx.doi.org/ Study 6 (Flight study -- also known as part C of Astro-Palate). Objectives: 1) to determine whether eating foods thought to be capable of improving mood will improve moods of astronauts in space when they are in a negative mood and 2) to determine whether eating such foods immediately prior to situations generating negative mood will ameliorate that mood. Crewmembers will complete questionnaires before and after specific snacks and tasks. We have received all necessary approvals for this study; our first subject reached the ISS in 2014. Next Steps. We will continue to recruit astronaut participants for this study and will continue working with the ISSMP team to implement the studies. Focus 3: The role of perceived variety in meal satisfaction, consumption, and mood. Goals. The goal of these studies was to curb boredom with repeatedly eaten foods by increasing appreciation for variety in one's diet without changing the available food items. Methods. The study had three steps: Step I screened participants; Step II oriented participants to the

	procedure; and Step III tested whether recalling variety reduces satiety. Step I: Participants rated their enjoyment of a granola bar and their willingness to eat these bars. They also indicated if they had a smart phone and were willing to complete 2 short surveys per day for 14 consecutive days on their smart phone. Participants who liked and were willing to eat granola bars, had a smart phone, and were willing to complete online surveys were invited to participate in the study. Step II: Participants attended an orientation session and were asked to select a single time of day during which they could eat a granola bar for the next 14 days. Participants were told to expect two text messages per day that contained a link to an online survey, and participants practiced accessing and responding to an example survey. Before leaving, participants were given a bag containing 14 granola bars. Step III: Step III took place on each of the 14 days following Step II. Participants received two text messages per day. Each morning participants received the first survey via text message. The survey was completed when the granola bar was to be eaten (i.e., the time specified during Step II) and measured satiation with the granola bars (e.g., desire for more, satisfaction). Every second day, prior to eating the granola bar and completing the satiation measures, participants randomly assigned to the control condition thought about the different musical artists they had listened to recently, whereas participants assigned to the experimental condition thought about the different foods they had eaten recently. In the evening, participants received a second text message, which contained a survey that measured the satisfaction with their diet in general. Results. As expected, liking of the granola bars decreased over time. On Day 14, participants that were reminded to think about the variety of foods they had eaten recently displayed less satiation than those in the control condition.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Articles in Peer-reviewed Journals	Wagner HS, Ahlstrom B, Redden JP, Vickers Z, Mann T. "The myth of comfort food." Health Psychology. 2014 Dec;33(12):1552-7. Epub 2014 Aug 18. http://dx.doi.org/10.1037/hea0000068 ; PubMed PMID: 25133833 , Dec-2014
Articles in Peer-reviewed Journals	Osdoba K, Mann T, Redden J, Vickers Z. "Using food to reduce stress: Effects of choosing meal components and preparing a meal." Food Quality and Preference. 2015 Jan;39:241-50. http://dx.doi.org/10.1016/j.foodqual.2014.08.001 , Jan-2015